

Multi-Evidence Natural Language Inference for Clinical Trial Reports

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1 Introduction

Clinical trials are research studies performed to test the efficacy and safety of novel treatments, and they are indispensable for the progression of experimental medicine. Clinical trial reports (CTRs) are documents that detail the methodology and results of a particular trial. Clinical professionals use the information in these CTRs to design personalized and targeted interventions, aligning treatment methodologies with the medical records of each patient and specific health conditions. However, there are over 400,000 CTRs available, being continually published at an ever increasing rate (World Health Organization 2024); It has become unfeasible to manually carry out comprehensive evaluations of all relevant literature when designing treatment protocols. Natural Language Inference (NLI) is a technique designed to assess the relationship between a given statement (hypothesis) and a text segment (premise), determining whether the premise entails, contradicts, or is neutral with respect to the hypothesis (Bowman et al. 2015). In the context of clinical trials, NLI can be employed to systematically analyze CTR, facilitating the identification of relevant studies that support or contradict specific clinical hypotheses. It presents a potential solution for large-scale interpretation and retrieval of medical evidence, allowing medical professionals to efficiently identify relevant CTRs to support personalized patient care (Jullien et al. 2023).

The NLI4CT data set is introduced in the paper by Jullien et al. (2023), titled "*NLI4CT: Multi-Evidence Natural Language Inference for Clinical Trial Reports*". The dataset comprises 2,400 expert-annotated entailment relations, with associated CTRs from ClinicalTrials.gov2, labels, and extracted evidence lists. The paper introduces two tasks, which are inference classification (determining whether a given statement is entailed or contradicted by the information in one or two CTRs) and evidence retrieval (identifying specific sentences within the CTR(s) that support the entailment decision made in Task 1). Baseline evaluations of eight state-of-the-art NLI models on task 1 revealed significant limitations, with a maximum F1 score of 0.644, highlighting the complexity of clinical inference tasks compared to general-domain NLI. For Task 2, several transformer-based ranking models and a traditional retrieval model were assessed, and the highest mean average precision (mAP) score of 0.786 was obtained, indicating that traditional retrieval methods can still be competitive in certain biomedical contexts. In my proposal, I present a comparison of diverse approaches to addressing these two tasks on the NLI4CT dataset, incorporating both general-purpose LLMs and NLI models.

2 Problem Statement

The exponential growth in CTRs presents a critical barrier to the advancement of precision medicine, where tailored therapeutic strategies depend on the rapid synthesis of high-quality

evidence. Over 400,000 CTRs are currently accessible globally, with an estimated 15% annual increase in publications as research output escalates (World Health Organization 2024). This deluge of data overwhelms healthcare professionals, who must manually parse lengthy, technically complex documents often exceeding 100 pages per trial to identify actionable insights. The challenge is compounded by the heterogeneous nature of CTRs, which vary in design, outcome measures, and patient cohorts, complicating cross-study comparisons (Ioannidis et al. 2014). Clinicians face significant cognitive overload when reconciling conflicting results, redundant findings, or nuanced subgroup analyses, leading to delays in decision-making and potential oversights in patient care (Bastian et al. 2010).

Systematic reviews are a common approach of evidence-based medicine, offering a structured, transparent, and reproducible approach to synthesize findings from multiple studies (Garcia-Alamino & Lopez-Cano 2020). However, Only a small minority of trial reports are incorporated into up-to-date systematic reviews, leaving most evidence unanalyzed or outdated (Bastian et al. 2010). This disconnect between evidence availability and practical utility is particularly acute for rare diseases or complex comorbidities, where CTRs are sparse yet clinically consequential. Current systematic review methodologies, which require months to synthesize evidence, are ill-suited to dynamic clinical environments where timely interventions are lifesaving (Higgins et al. 2019). Addressing these challenges demands scalable solutions that automate evidence synthesis, prioritize high-impact findings, and reconcile contradictory results across CTRs accurately.

3 Research Questions

This study seeks to investigate the application of Natural Language Inference (NLI) techniques to inference validation and automate evidence retrieval in CTRs. The following research questions are formulated to systematically evaluate the effectiveness of diverse methodologies, model architecture, and domain-specific adaptations:

- **Zero-Shot Performance of Generative LLM vs. Trained NLI models:** Does a general-purpose generative LLM (Deepseek) achieve higher classification accuracy in determining entailment or contradiction relations between hypotheses and CTRs compared to task-specific BERT-based models (BERT, BioBERT) fine-tuned on the NLI4CT dataset?
- **Effectiveness of LLMs and Embedding Model Combinations in a Naïve RAG Framework:** Which combination of LLM (Llama, T5, Mistral, DeepSeek) and embedding model (Linq-Embed-Mistral) yields the highest performance in a naïve RAG framework, as measured by accuracy for inference classification and mean average precision (mAP) for evidence retrieval?
- **Impact of Domain-Specific LLMs on Task Performance:** Do domain-adapted LLMs such as BioMistral-7B, Bio-ClinicalBERT, LLaMA3-OpenBioLLM-8B, and BioGPT offer improved performance over general-purpose models in both inference classification and evidence retrieval tasks?
- **Enhancing RAG with Advanced Prompting and Retrieval Techniques:** Do methodological advancements such as Chain-of-Thought (CoT) prompting, task-specific prompt engineering, and retrieval reranking improve the performance of naïve RAG frameworks in evidence retrieval and inference classification?

Together, these research questions aim to identify the optimal combination of model design, adaptation strategies, and inference mechanisms for robust, scalable analysis of CTRs to support evidence-based, personalized medical decision-making.

4 Methods

This study adopts an experimental research design to evaluate and compare the performance of multiple language models and retrieval frameworks in the context of NLI and evidence retrieval from CTRs. The methodology is organized around four core research questions, focusing on model performance, domain adaptation, and optimization.

4.1 Dataset

The primary dataset used in this study is the NLI4CT dataset, a publicly available benchmark specifically designed for natural language inference tasks within the biomedical domain. This dataset contains pairs of hypotheses (representing patient-specific medical statements) and premises (extracted segments from CTRs), labeled as entailment or contradiction. The CTRs are structured into four sections: Eligibility Criteria, which outlines the conditions patients must meet to participate in the clinical trial; Intervention, detailing the type, dosage, frequency, and duration of the treatments being studied; Results, which includes the number of participants, outcome measures, units, and trial outcomes; and Adverse Events, which documents observed signs and symptoms related to the trial (Jullien et al. 2023).

4.1.1 Zero-Shot vs Fine-Tuned NLI Models

To compare general-purpose and task-specific models, zero-shot inference will be conducted using DeepSeek, a generative LLM, without domain adaptation. This will be compared against BERT and BioBERT models fine-tuned on the NLI4CT dataset. The primary evaluation metric is classification accuracy for entailment versus contradiction labels.

4.1.2 Retrieval-Augmented Generation (Naïve RAG)

A naïve RAG framework will be implemented by combining embedding-based retrieval using Linq-Embed-Mistral with generative LLMs such as LLaMA, T5, Mistral, and DeepSeek. Retrieved segments will be appended to the input prompt for classification and compared with the ground truth evidence of the hypothesis. The evaluation will include:

- **Accuracy** for NLI classification
- **mean average precision (mAP)** for evidence retrieval and matching

4.1.3 Domain-Specific LLMs

To assess the effect of domain adaptation, models such as BioMistral-7B, LLaMA3-OpenBioLLM-8B, BioGPT, and Bio-ClinicalBERT will be evaluated for inference classification, and the results will be compared with the results obtained previously from general-purpose models.

4.1.4 Prompting and Retrieval Enhancements

As an optimization step, several enhancements will be implemented to refine the classification accuracy and the evidence retrieval efficiency. Chain-of-Thought (CoT) prompting and

task-specific prompt engineering will be used to improve the inference classification. Different retrieval reranking approaches (bi-encoder and cross-encoder) will be used to improve the quality of retrieved evidence.

4.2 Evaluation Metrics

The evaluation of model performance is based on standard metrics suitable for each task. Entailment classification is assessed using accuracy, precision, recall, and F1-score. For evidence retrieval, mean average precision (mAP) is employed.

Task	Metrics
Entailment Classification	Accuracy, Precision, Recall, F1
Evidence Retrieval	Mean Average Precision (mAP)

Table 1: Evaluation metrics used for each task.

5 Ethical Implications

The nature of the medical domain raises several ethical considerations that must be addressed to ensure responsible and compliant AI development. This research prioritizes data privacy and anonymity by using the publicly available NLI4CT dataset to comply with regulations like GDPR and HIPAA, which is ensured through exclusion of patient identifiers and sensitive health information. Additionally, the study emphasizes explainability and transparency, recognizing that clinical decision support systems must provide interpretable outputs. Models will be evaluated not only on performance metrics but also on their ability to obtain evidence for the decision they made.

The work is strictly confined to research purposes and does not endorse direct clinical use without expert oversight and validation. Compliance with institutional ethical guidelines, potential Institutional Review Board (IRB) approvals, and adherence to licensing terms for external models further reinforce responsible practices. By integrating these safeguards, the research aims to maintain a commitment to ethical standards and societal trust.

6 Requirements and Feasibility

Functional Requirements

- **Entailment Classification Modules:** Classify hypothesis–CTR pairs into Entailment or Contradiction using general-purpose LLM (Deep seek) and task-specific fine-tuned models (BERT, BioBERT).
- **RAG Framework integration:** Integrate different LLMs such as LLaMA, T5, Mistral, DeepSeek with an embedding model in a naïve RAG framework and compare both inference accuracy and evidence retrieval efficiency.
- **Integrate Domain specific LLMs:** Integrate domain specific LLMs such as BioMistral-7B, LLaMA3-OpenBioLLM-8B, BioGPT, Bio-ClinicalBERT and compare both inference accuracy and evidence retrieval efficiency.
- **Advanced Prompting Techniques:** Implement CoT, retrieval reranking, and prompt engineering to assess their impact on task performance.

- **Statistical Framework:** Develop the statistical analysis module to systematically compare the results obtained from the different methods and architectures.

Non-Functional Requirements

- **Performance:** Models must achieve classification accuracy and F1-score above baseline metrics reported in NLI4CT benchmarks.
- **Scalability:** The solution must be able to handle multiple CTRs and hypotheses and adapt to extended medical datasets.
- **Usability:** Output (classification + evidence) must be human-readable, with reasoning traces where applicable.
- **Security and Compliance:** All data sets and tools used must comply with ethical guidelines and licensing requirements.
- **Reproducibility:** Experiments must be documented and easily reproducible. (Maintained a well-structured GitHub repo)

User Requirements

- **Clinical practitioner:** Require accurate, interpretable models for medical NLI tasks.
- **Academic Evaluators:** Expect clear methodology, reproducible results, and a well documented pipeline for evaluation.

Technical Feasibility

The project leverages accessible, open-source libraries such as Hugging Face Transformers, PyTorch, and LangChain. All proposed models (DeepSeek, BioGPT, BERT, BioBERT, LLaMA3-OpenBioLLM) are available for academic use and the NLI4CT dataset is publicly accessible. The project can be developed using local GPU/CPU for initial testing. For larger models, API access or lightweight quantized versions will be used to reduce resource constraints.

Risk Assessment

Potential risks associated with the experimental design and model implementation are identified and addressed through appropriate mitigation strategies, as summarized in Table 2.

Risk	Impact	Mitigation Strategy
Fine-tuning fails	Medium	Use pre-trained models
High computing cost for large models	High	Use quantized versions or access models via APIs
Delays in implementation or unexpected bugs	Medium	Use agile development with targeted milestones

Table 2: Identified risks, their impact levels, and mitigation strategies.

7 Project Plan

The following diagram illustrates the overall project timeline and key milestones.

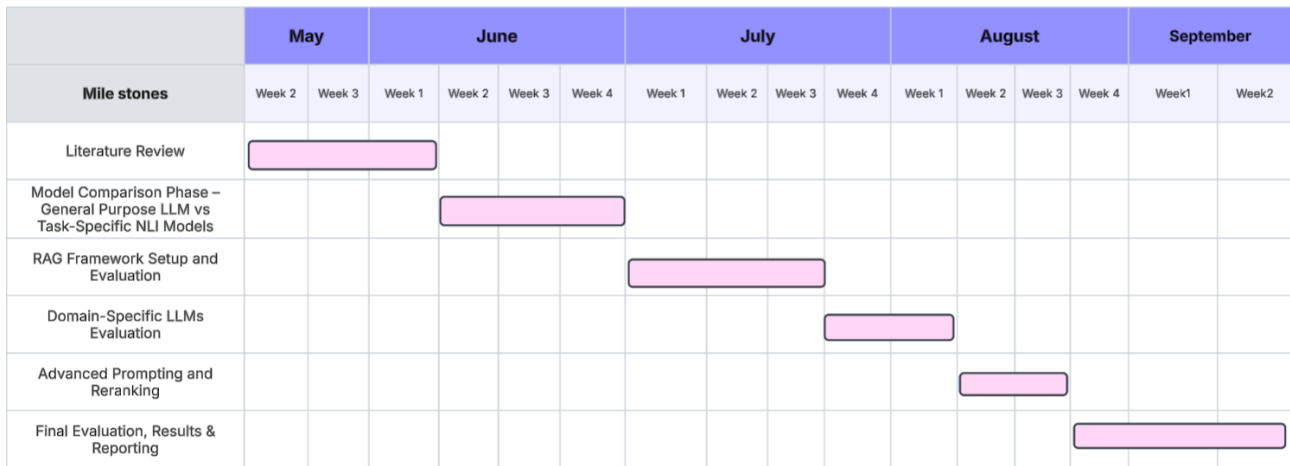


Figure 1: Project plan timeline with major milestones.

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